



Indian Institute of Information Technology Surat
B.Tech-II Semester IV Computer Science and Engineering
End Sem Examination, Design Analysis and Algorithm (CS 405)

Marks: 100

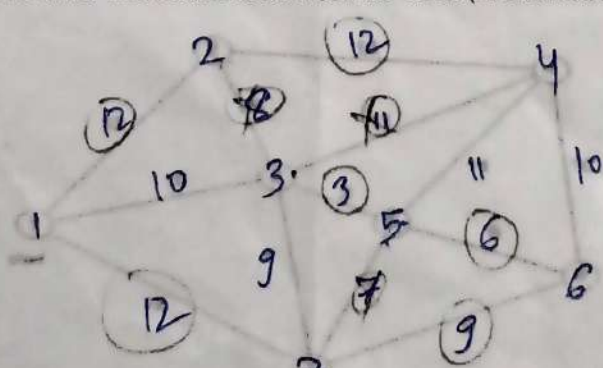
Date: 16th May 2025

Time: 10:00 AM to 01:00 PM

Instructions:

1. Attempt all questions

2. Make necessary assumptions wherever required.

[1]	(a) Can the master method be applied to the recurrence: $T(n)=4T(n/3)+n^2 \log n$? Why or Why not? Give an asymptotic upper bound for this recurrence. [3] (b) Design an algorithm to find the sum of smallest $\log_2 n$ elements in an unsorted array of n distinct elements. [5] (c) Find the time complexity for the following: [2] <pre>void linearLog(int n) { for (int i = 0; i < n; i++) { for (int j = 1; j < n; j *= 2) { // do something } } }</pre>															
[2]	Write an algorithm to compute the length of Longest Common Subsequence (LCS). Determine an LCS of $\langle A,B,C,D,B,A,C,D,F \rangle$ and $\langle C,B,A,F \rangle$. [10]															
[3]	(a) Given 6 activities along with their start and end times as: $S = (A_1,A_2,A_3,A_4,A_5,A_6)$, $S_i = (1,3,0,5,8,5)$, $F_i = (4,5,6,7,9,9)$. Compute an execution schedule having the maximum number of non- conflicting activities. [5] (b) Given the following characters and their frequencies, compute the Huffman codes for each character: [5] <table border="1" data-bbox="648 1024 953 1274"><thead><tr><th>Character</th><th>Frequency</th></tr></thead><tbody><tr><td>A</td><td>5</td></tr><tr><td>B</td><td>9</td></tr><tr><td>C</td><td>12</td></tr><tr><td>D</td><td>13</td></tr><tr><td>E</td><td>16</td></tr><tr><td>F</td><td>45</td></tr></tbody></table>	Character	Frequency	A	5	B	9	C	12	D	13	E	16	F	45	
Character	Frequency															
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E	16															
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[4]	(a) A salesman needs to visit 7 cities and return to the starting point. The goal is to find the shortest possible route that visits each city exactly once and returns to the origin. Find the shortest route that starts and ends at the same city (e.g., 1), visiting all cities once. The distances between the cities (in kilometres) are given as: [5]  (b) Write the algorithm for the Hamiltonian Cycle. Does the above figure [i.e. in Q.4(a)] follow Hamiltonian Cycle, If Yes, justify your answer. [5]															

[5]	(a) Given a set of integers: $S = \{3, 34, 4, 12, 5, 2\}$ and a target sum $T = 9$, Determine whether there exists a subset of S whose sum is exactly 9. If yes, provide one such subset using backtracking. (b) Consider the knapsack instance $n=3$, $M=5$, $(W_1, W_2, W_3) = (2, 3, 4)$ and $(P_1, P_2, P_3) = (2, 3, 5)$. [Where, Weight = $(W_1, W_2, W_3, \dots, W_n)$, Profits = $(P_1, P_2, P_3, \dots, P_n)$, n = Number of Elements, M = Maximum Capacity]. FRACTIONAL	[5] [5]
[6]	(a) Write the algorithm to find all-pair shortest path. For the graph (directed, weighted), apply Floyd-Warshall algorithm for constructing the shortest path. <pre> graph TD 0((0)) -- 4 --> 1((1)) 0((0)) -- -2 --> 2((2)) 1((1)) -- 3 --> 2((2)) 1((1)) -- -1 --> 3((3)) 3((3)) -- 2 --> 2((2)) </pre> (b) Write the Algorithm of Bellman-Ford. Also, mention the advantages, disadvantages and its time complexity.	[10] [10]
[7]	(a) Recall that $a \mid b$ if there exists an integer k such that $ak = b$. Prove that if $a \mid b$ and $c \mid d$, then $ac \mid bd$. (b) Use Euclid's algorithm to find the greatest common divisor of 155 and 93. Then, express this greatest common divisor as an integer linear combination of 155 and 93.	[5] [5]
[8]	(a) In an RSA cryptosystem, Alice's public key is $(n = 33, e = 7)$. Suppose Bob wishes to encrypt the message $M = 2$. Compute the ciphertext that Bob sends to Alice over an insecure channel. (b) In an RSA cryptosystem, suppose Alice chooses $n = 33$, where $n = p \cdot q$ with $p = 3$ and $q = 11$. She selects her public exponent as $e = 7$. Compute the corresponding decryption key d, i.e., the modular inverse of e modulo $\phi(n)$.	[5] [5]
[9]	(a) Determine whether the complete bipartite graph $K_{3,4}$ contains a Hamiltonian cycle. Justify your answer. Determine whether the complete bipartite graph $K_{4,4}$ contains an Eulerian cycle. Justify your answer. What is the complexity of the problem of determining whether a graph contains a Hamiltonian cycle? (b) The clique problem asks to determine, given an input graph G and a positive integer k, whether G contains a set of k vertices that are pairwise adjacent. The independent set problem asks to determine, given an input graph G and a positive integer m, whether G contains a set of m vertices that are pairwise nonadjacent. Assuming the clique problem is NP-complete, give a polynomial time reduction that would prove that the independent set problem is NP-complete.	[6] [4]

END